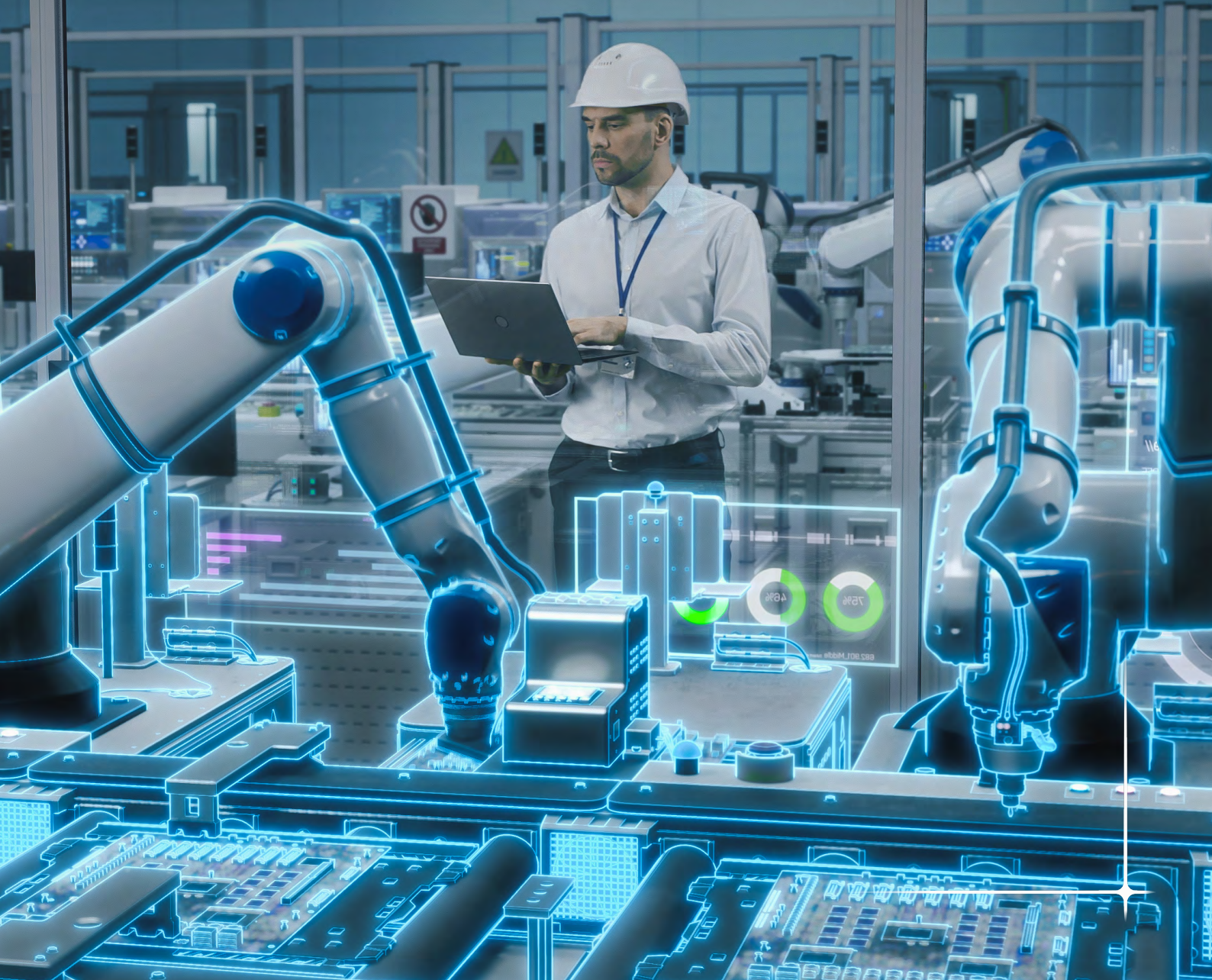




How the Factory of the Future Is Reshaping the Economics of Manufacturing Competitiveness

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AI is reshaping the economics of manufacturing faster than many leaders realize. CEOs must act quickly or they risk falling behind those who grasp the new logic of competition.

Across sectors, AI-enabled production setups are changing conversion costs and unlocking productivity savings of up to 60%. And the stakes extend beyond the factory floor: roughly \$1.03 trillion of manufacturing value is at risk of relocation out of Western Europe and the Nordics, with another \$440 billion at risk in the United States. Against this backdrop, CEOs may be tempted to upgrade their existing factories immediately. That decision, however, isn't always that straightforward—and a key question looms.

The results of our proprietary quantitative analysis, combined with a global survey of 1,000 manufacturers, reveal the tension. In some sectors, upgrading to Factory of the Future (FoF) capabilities in a high-cost country can be a more competitive option than offshoring, even if lower-cost countries also upgrade. But for other sectors, relocation remains the best strategy. To assess these choices, we developed a Manufacturing Competitiveness Index—a composite tool integrating 42 cost and qualitative factors.

Applying this index reveals highly differentiated outcomes: competitive advantage, parity, or a lingering gap. A food manufacturer based in Western Europe adopting FoF to serve its domestic market, for example, would be more competitive than relocating to China, securing a 14 percentage point cost advantage. Meanwhile, an engineered components manufacturer can reach competitiveness parity by upgrading its factories. And in electronics, relocation to China generally remains the better choice under pure cost considerations: a FoF-upgraded factory still faces a 15 percentage point cost gap.

Such scenarios represent a seismic shift, challenging the logic on which global manufacturing footprints have been built. The critical variable is no longer relative labor costs and logistics from suppliers and to customers, but how effectively a facility can be transformed into a highly productive Factory of the Future. Manufacturing CEOs charting the course ahead need to consider six dimensions:

- Strategic importance of localization
- Degree to which FoF can alter the cost position
- Ability to ultimately realize productivity gains
- Impact of sector-specific tariffs
- Local readiness, particularly in areas such as digital infrastructure and workforce capabilities
- Broader business context, especially whether it is a brownfield or a greenfield project

CEOs must determine not only where to produce in existing cost structures, but where the Factory of the Future can be deployed in ways that fundamentally shift cost structures and performance outcomes.

AI Transforms the Factory Floor

Manufacturers have aspired to build the Factory of the Future for years. What's changed is that three converging breakthroughs—agentic systems, AI (both physical and virtual AI), and computing power enabling performant simulations—have made end-to-end production redesign economically viable at scale. (See sidebar, “Three Breakthroughs.”)

The FoF holistically transforms production. It creates value through processes that have been redesigned from end to end, going beyond isolated automation or AI-driven use cases. This involves the orchestration of multiple capabilities into a coherent, integrated redesign to optimize how materials move, machines interact, decisions are made, and variability is managed. Improvements in one part of the factory reinforce gains in others, creating a compounding effect across the entire operation.

Rethinking the Economics of Production

In traditional settings, productivity improvements tend to be incremental and localized. Manufacturers upgrade machines and automate isolated cells, but they don't redesign entire workflows, so they don't realize their full potential.

The FoF operates differently. Because the entire production setup is holistically redesigned, multiple cost drivers shift simultaneously. Automation reduces labor intensity while enabling more stable processes that improve yield and reduce waste. AI-driven control systems optimize energy use and equipment performance

continuously. Integrated planning and orchestration reduce idle time, smooth material flows, and compress working capital. A new logic for production costs emerges.

This effect is visible across industries. Here, we've selected three composite examples to illustrate the impact.¹

- **Example #1: a coffee roasting and packaging company, Germany.** Combining automation, intelligent process and parameter control, predictive maintenance, and centralized data coordination can reduce labor requirements while also optimizing energy consumption and warehousing costs. This translates into labor savings of nearly 60% and overall conversion costs falling by more than 40 percentage points.
- **Example #2: a pharmaceutical company, US.** In tablet and capsule manufacturing, the FoF integrates digital process control (for instance, digital twins for batch scheduling and recipe optimization), connected operations monitoring (IoT-enabled deviation tracking), advanced analytics (for yield improvement and root-cause analysis), and digital quality and release (real-time in-process quality control and accelerated batch release). Together, these changes reduce labor by more than 60% and lower total conversion costs by 30 percentage points.
- **Example #3: a battery cell manufacturing company, South Korea.** Process redesign—such as continuous mixing, optimized coating, and AI-enabled quality control—reduces labor requirements by approximately 30%. Redesign across coating, calendaring, stacking, formation, and quality control lowers conversion costs by more than 25 percentage points. (See Exhibit 1.)

1. These composite examples are grounded in BCG case work, other company examples, and BCG Factory Advisor, the firm's proprietary tool on factory redesign.

Three Breakthroughs

For years, manufacturers have automated isolated tasks. Three converging breakthroughs have now made it possible—and economically viable—to transform how entire factories operate at scale:

The Rise of Agentic Systems. The multiplication and diversification of agentic solutions is accelerating rapidly. This explosion of available solutions now offers manufacturers the ability to orchestrate complex, end-to-end redesigns of production systems in ways that were previously out of reach.

The Maturity of Physical and Virtual AI. Both physical and virtual AI have reached an inflection point. Supported by foundation models and a narrowing simulation-to-reality gap, physical AI has cut robot training time by 70% while expanding automatable work by 50%. Virtual AI, meanwhile, is now deployed at scale across factory operations—enabling self-controlling machines, predictive maintenance, and intelligent production scheduling.

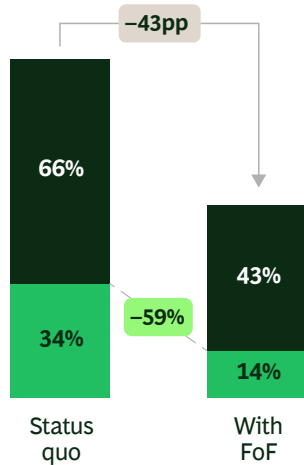
The Expansion of Computing Power. A thousandfold increase in computing power over the past decade dramatically expands the ability to run advanced analytics, simulations, and AI models at scale and at lower cost, making system-level production redesign economically viable.

EXHIBIT 1

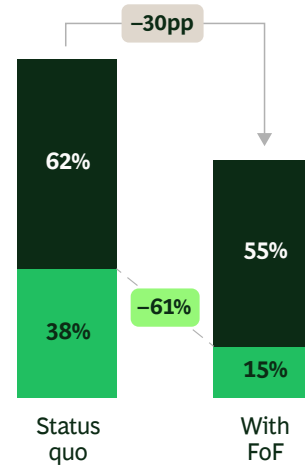
How the Factory of the Future Significantly Lowers Conversion Costs

Composite examples¹

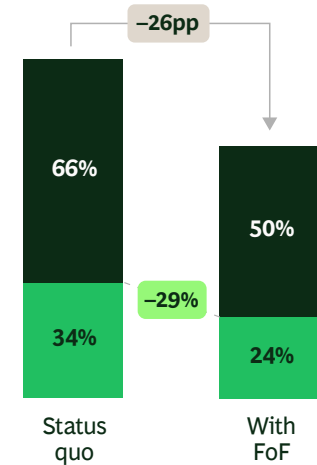
Coffee roasting and packaging company
Germany



Pharmaceutical company
US



Battery cell manufacturing company
South Korea



■ Labor ■ Other conversion costs ■ Impact on conversion cost ■ Labor savings

Source: BCG Institute Manufacturing Competitiveness Index.

Note: FoF = Factory of the Future. Other conversion costs include cost of energy, depreciation costs, and other operational expenditures (excluding tariffs and logistics). The status quo is set to a base of 100.

¹These composite examples are grounded in BCG case work, other company examples, and BCG Factory Advisor, the firm's proprietary tool on factory redesign.

Across these scenarios, significant labor savings are reinforced by gains in energy, materials, yield, and throughput. The result is a structural shift in how value is created within the factory.

The Six Dimensions Changing the Map of Manufacturing

As conversion costs shift, so does the relative competitiveness of locations. The question is no longer where labor is cheapest, but whether upgrading to a Factory of the Future outperforms relocating to a lower-cost site. Answering it requires assessing six dimensions in sequence: starting with resilience needs, then moving to cost impact, qualitative readiness, and broader business context.

1. Localization Strategy. When weighing footprint decisions, manufacturers must consider whether their sector and customer base demand proximity and higher resilience. Increasingly, the answer is yes.

As geopolitical uncertainty deepens and supply chain volatility becomes a structural risk, manufacturers have increasingly built resilience by producing where they sell. The shift is well underway: a 2025 BCG Institute survey of 1,000 manufacturing executives found that supply chain disruptions and geopolitical risks now both rank among manufacturers' top five challenges.

This pattern is visible across geographies and sectors. In the US, regionalization is taking the greatest hold in semiconductors and EV batteries, driven by shifting policies. In Europe, the same logic is at work in energy-intensive industries, such as steel, chemicals, and metals, motivated by the EU's Clean Industrial Deal that seeks to reduce dependence on non-European suppliers.

2. FoF Impact on Local Operating Costs. The FoF does not benefit all locations equally—its impact depends on the sector and on local cost factors such as energy, labor, and materials. The impact of the FoF is felt more strongly in high-cost regions: by automating labor-intensive tasks, optimizing energy consumption, and improving yield and throughput, FoF narrows the cost gap between high-cost and low-cost locations. Critically, this effect is larger in absolute terms where costs are highest to begin with.

How much the gap narrows (or is even reversed entirely) depends heavily on the sector. Two factors are most influential. The first is automation potential and its resulting impact on costs, which varies significantly across sectors. The second is the share of logistics costs. In sectors where this share is significant, such as food and beverages, proximity to end markets amplifies the FoF's compression effect.

Where both factors are favorable, the combination can be enough not just to narrow the gap with low-cost locations but to reverse it entirely. In food processing, we find that Germany can reach a cost advantage of up to 14 percentage points serving its domestic market. But in other sectors, such as battery cell manufacturing, a meaningful 15 percentage point gap remains even with full FoF deployment. (See Exhibit 2.)

3. Local Ability to Realize Productivity Gains. Beyond its impact on operating costs, the case for investing in the FoF varies significantly across locations. Because automation reduces labor costs in absolute terms rather than

proportionally, each unit of labor removed generates greater absolute savings in high-wage countries, shortening payback periods—the time to recoup a manufacturer's initial FoF investment—considerably. (See Exhibit 3.)

In the automotive industry, for example, payback periods are materially shorter in high-cost regions. In modeled cases where the US serves as the baseline, other high-cost countries remain in a broadly comparable range, while similar deployments in lower-cost markets can require payback periods that are roughly two-to-eight times longer.

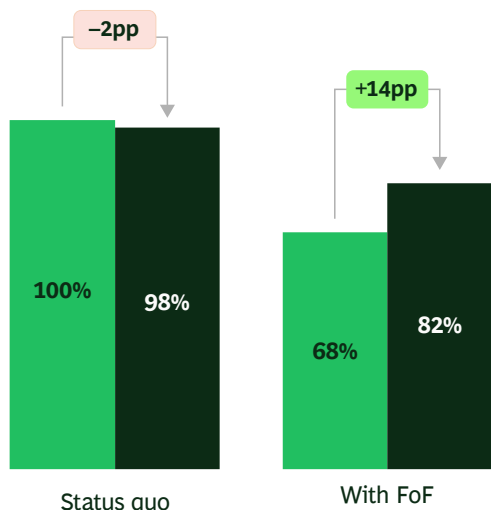
However, that does not mean the highest-wage locations always offer the strongest economics. In parts of Europe, higher labor restructuring costs—such as severance, notice periods, and implementation friction—can offset part of that advantage relative to the US. Companies hesitate to invest because reducing labor triggers real, immediate costs that offset the savings.

EXHIBIT 2

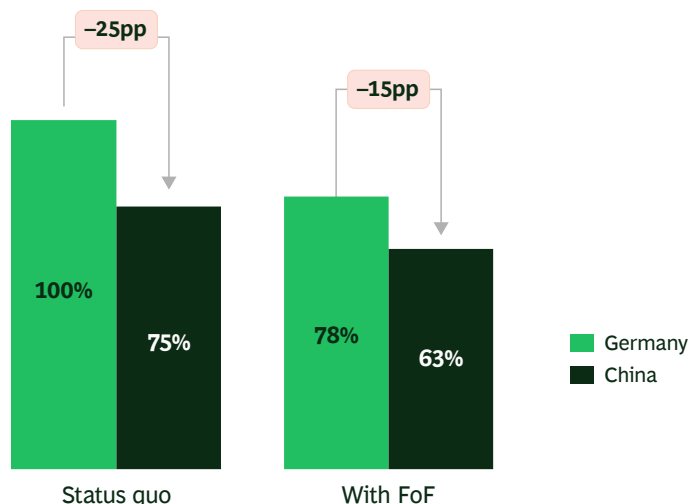
FoF Impact Is Greater in High-Cost Regions but Varies by Sector

FoF impact on production costs in China and Germany to serve the German market

Food processing



Battery cell



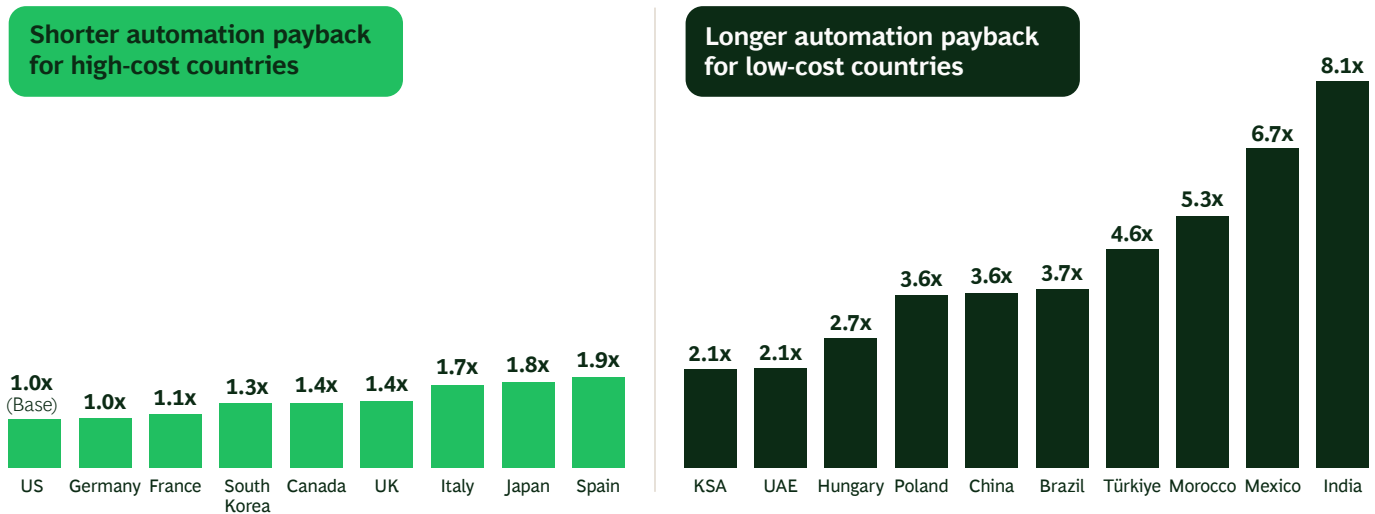
Source: BCG Institute analysis.

Note: FoF = Factory of the Future. Cost includes conversion costs (cost of energy, labor, other operational expenditures, depreciation costs), logistics costs (inbound and outbound) and tariffs, with FoF impact adjusted to local labor productivity enhancement optimum for each country.

EXHIBIT 3

Measuring the Return on Factory of the Future Investment

Relative payback period for a Factory of the Future automation in automotive



Sources: EIU; UNIDO; ILO; national labor institutions; BCG Institute analysis.

Note: FoF = Factory of the Future. Factory of the Future automation includes the following: automation of assembly, piling palletizing, and loading and unloading of input boxes, with payback period defined as the total number of years for annual net savings to recoup initial investment (considering FoF investment required, people restructuring costs with severance, notice, and implementation period). Countries' relative payback is expressed in comparison to the US.

4. The Role of Sectoral Tariffs. Trade policy can fundamentally change the manufacturing cost equation. By raising the cost of imported goods, tariffs can swiftly wipe out the economics of offshoring. Our 2025 global manufacturing survey revealed that a 25% tariff rate is enough to break the export business case for 90% of manufacturers. In that sense, tariffs can accelerate localization decisions.

Yet tariff protection is not a substitute for improving competitiveness. It may delay the pressure to invest in advanced production setups, but it does not remove it. Manufacturers that rely on protection without upgrading their operations face a compounding risk: production costs remain structurally higher than those of peers investing in the FoF elsewhere, eroding long-term competitiveness against markets that continue to modernize. And if trade policy conditions shift, that underlying cost disadvantage is suddenly exposed, with little time to respond.

5. Local Qualitative Readiness. Beyond cost, leaders should consider how site-selection decisions are shaped by qualitative factors, including the political and business environment, talent availability, infrastructure, and market and supply-chain depth. Some qualitative factors become even more important in the context of the FoF. In our

global manufacturing survey, 87% of respondents indicated that access to talent and skills becomes more critical to sustaining a FoF deployment. And 69% said the same of infrastructure—with digital infrastructure ranking as the most critical component within infrastructure, reflecting the importance of computing power in AI-heavy processes. Without that foundation, technological potential does not translate into competitive advantage.

To assess countries' qualitative readiness for FoF deployment, we measured workforce skills and digital infrastructure readiness for each country on a scale from 1 (very low) to 10 (very high). The results reveal a clear pattern: the US leads, benefiting from strong tech infrastructure and a deep talent base, with Germany close behind as the top European performer. East Asian economies (Japan, South Korea) follow, driven by strong telecommunication networks and a skilled workforce. Western European economies, Canada, and China come next, with China's position notably supported by the scale of its STEM graduate pool. Countries such as Mexico, India, and Morocco score lower today—reflecting a meaningful improvement opportunity for those that invest in workforce capabilities and digital infrastructure. (See Exhibit 4.)

EXHIBIT 4

Country Ranking On Skills and Digital Infrastructure Readiness

SCORING FROM 1 (VERY LOW) TO 10 (VERY HIGH)

Ranking		Total		Skills		Digital infrastructure	
1	US	8.4		9.0		7.9	
2	Germany	8.2		9.0		7.4	
3	Japan	7.6		8.0		7.1	
4	South Korea	7.6		8.0		7.1	
5	UK	7.5		9.0		6.0	
6	Canada	7.2		8.0		6.4	
7	France	7.1		7.0		7.1	
8	China	6.6		7.0		6.1	
9	Italy	6.3		6.0		6.6	
10	Poland	5.9		7.0		4.9	
11	Spain	5.8		5.0		6.6	
12	Hungary	5.4		6.0		4.9	
13	Brazil	5.0		6.0		4.0	
14	UAE	5.0		6.0		4.0	
15	India	5.0		7.0		3.0	
16	Mexico	4.6		6.0		3.3	
17	Türkiye	4.1		5.0		3.1	
18	KSA	4.0		4.0		4.0	
19	Morocco	3.1		4.0		2.3	

Sources: UNESCO; OECD; national educational statistics; Portulans Institute; IFR; BCG Institute analysis.

Note: Composite index scoring from 1 to 10 based on the unweighted average scoring of two criteria: skills (based on mean years of schooling and yearly number of STEM graduates) and digital infrastructure (based on network readiness index measuring countries' digital maturity across technology, people and governance, and industrial robotic density deployment).

6. Business Context. This includes brownfield versus greenfield investment and other decisions that influence which production locations are most competitive. To capture how business context interacts with the five factors above, we built a Manufacturing Competitiveness Index—a composite scoring tool integrating 42 cost and qualitative factors that rates the competitiveness of any production location on a scale from 1 to 10, both without and with the impact of the Factory of the Future, across industrial sectors, customer market, asset, and investment type. (See sidebar, “Manufacturing Competitiveness Index Methodology.”)

In brownfield settings, existing assets, local capabilities, and relocation costs create stickiness that raises the threshold for moving production. To serve the German market, for example, upgrading an existing factory in Germany with the FoF can become a more competitive play than relocating to

China, but the outcomes vary greatly across sectors. Based on our Manufacturing Competitiveness Index, upgrading to the Factory of the Future is more competitive in food and beverages, reflecting the cost gap compression from FoF deployment and the weight of logistics costs. Yet the advantage varies within the sector, with food processing benefiting more than beverages, where automation is more mature. In aerospace and defense, competitiveness reaches parity. In electronics and electrical, however, structural competitiveness differences persist: even with full FoF deployment, a German factory remains less competitive than a relocation to China, reflecting deeper supply chain and cost disadvantage. (See Exhibit 5.)

Manufacturing Competitiveness Index Methodology



BCG Institute Manufacturing Competitiveness Index (MCI) scores the competitiveness of each production location on a scale from 1 (very low) to 10 (very high) for a specific business context defined by: industrial sector, end customer market, asset type (asset-light or asset-heavy), and investment type (brownfield or greenfield) for two scenarios: without the Factory of the Future and with the Factory of the Future.

Scope. Included are 54 countries representing more than 95% of world manufacturing value added, across 47 industrial sectors, two asset types, two investment types, and two scenarios (without and with the Factory of the Future).

Methodology. The competitiveness scoring is derived from the weighted average of 42 cost and qualitative factors, with weightings derived from the BCG Institute's global survey of 1,000 manufacturing executives and tailored by industry and business context.

Cost scoring is based on the cost competitiveness of each country depending on the business context considering 17 factors:

- Running cost (material, labor, energy, depreciation, other operational expenditures, inbound and outbound logistics)

- Tariffs (inbound and outbound tariffs)
- One-offs (cost of land, factory building, permitting, hiring and training, FoF upgrade, factory closure, people restructuring, and cost of capital)

Qualitative scoring is a weighted average of 25 factors across five clusters:

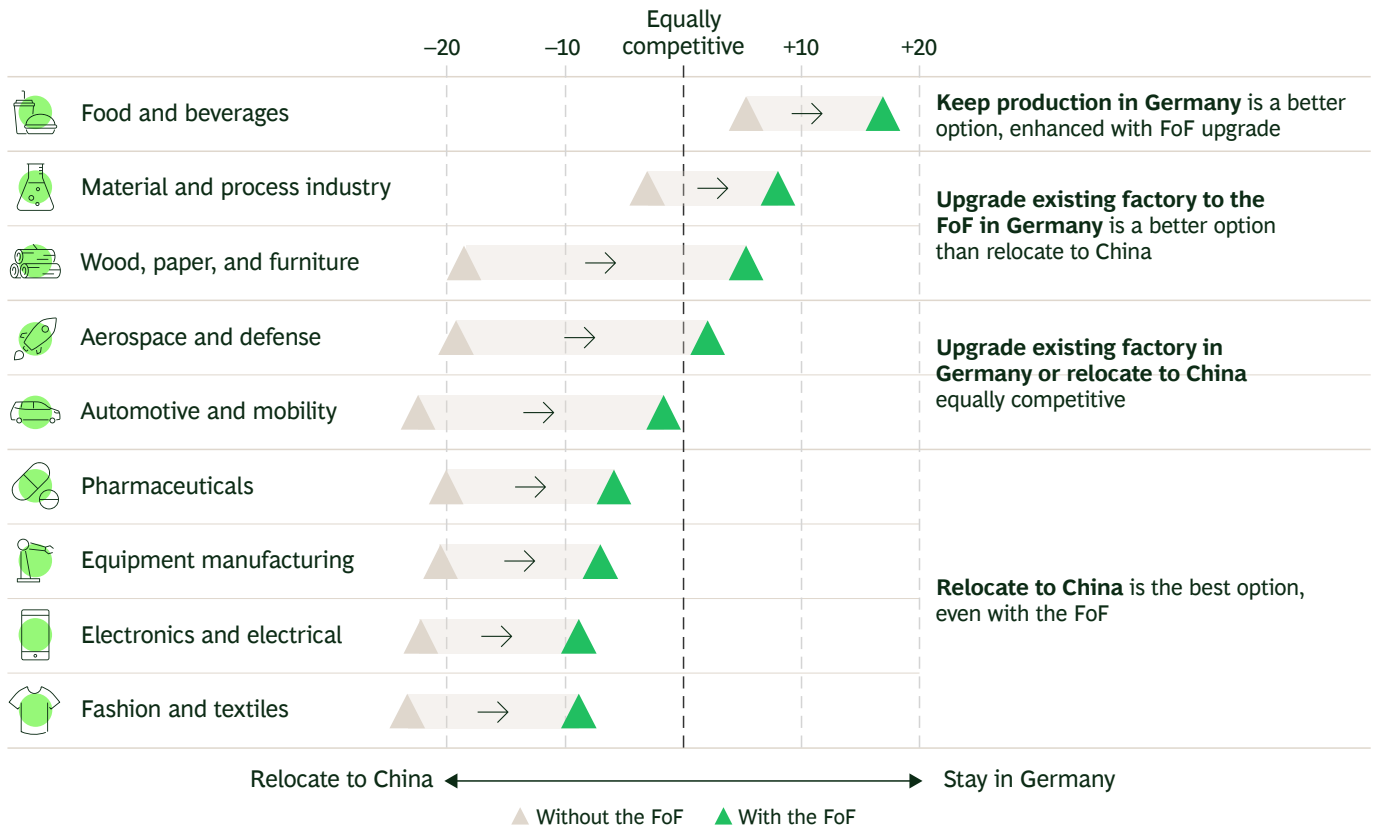
- Political environment (political stability, geopolitical risk, business readiness, level of corruption, rule of law)
- Business environment (past and future GDP growth, total GDP, financial stability, environmental performance)
- Talent and skills (labor freedom, labor shortages, human capital and skills, number of STEM graduates, livability)
- Infrastructure and utilities (access to electricity, access to water and utilities, digital and tech infrastructure, logistics providers, road conditions)
- Market and supply chain (market size, market growth, presence of competitors, availability of raw materials, quality of supplier ecosystem)

EXHIBIT 5

The FoF Can Tip the Scale in Many Sectors for Brownfield Projects

Germany vs. China for brownfield German factory serving local market

MANUFACTURING COMPETITIVENESS INDEX SCORING GAP (%)



Source: BCG Institute Manufacturing Competitiveness Index.

Note: FoF = Factory of the Future. Manufacturing Competitiveness Index rates on a scale from 1 (very low) to 10 (very high) the weighted combination of 42 cost and qualitative factors in the context of a brownfield German factory serving German market, weightings derived from a survey of 1,000 manufacturing executives.

The story is different in the US. In the automotive sector serving the domestic market, for instance, current tariff levels make relocation to China or Mexico uncompetitive. Yet, if tariffs were to be eased, US manufacturers would face a meaningful competitiveness disadvantage against these lower-cost locations. Upgrading to the FoF changes that equation: it would secure a durable long-term competitiveness advantage—independent of tariff conditions. (See Exhibit 6.)

In greenfield settings, by contrast, FoF capabilities can be embedded from the outset, making site selection more responsive to underlying differences in operating cost and initial one-offs (cost of land and factory building). Here, the Germany-versus-China comparison shows a differentiated picture as the FoF tips the scale in fewer sectors than in brownfield settings. Germany holds a competitive advantage in food and beverages serving the local market. However, in aerospace and defense as well as electronics and electrical, meaningful gaps remain in favor of China. (See Exhibit 7.)

Still, context matters: company-specific needs and desire for localization can alter the calculus for leaders, especially in defense and other sectors where region-for-region production is preferred.

EXHIBIT 6

The Factory of the Future Prevents Loss of Temporary Tariff Advantage

US vs. Mexico for brownfield US factory serving local market

MANUFACTURING COMPETITIVENESS INDEX SCORING GAP (%)



Source: BCG Institute Manufacturing Competitiveness Index.

Note: FoF = Factory of the Future. Manufacturing Competitiveness Index rates on a scale from 1 (very low) to 10 (very high) the weighted combination of 42 cost and qualitative factors in the context of a brownfield US factory serving domestic market, weightings derived from a survey of 1,000 manufacturing executives.

Reshaping Competition Across Regions

In reshaping footprint economics, the Factory of the Future is redrawing the competitive map of global manufacturing. The result is not a wholesale reversal of manufacturing competitiveness patterns, but instead a highly variable landscape. Regions that combine ability to effectively deploy the FoF and invest in their qualitative capabilities will gain a growing advantage, while traditional cost leaders will remain competitive primarily when automation potential is lower or slower to materialize.

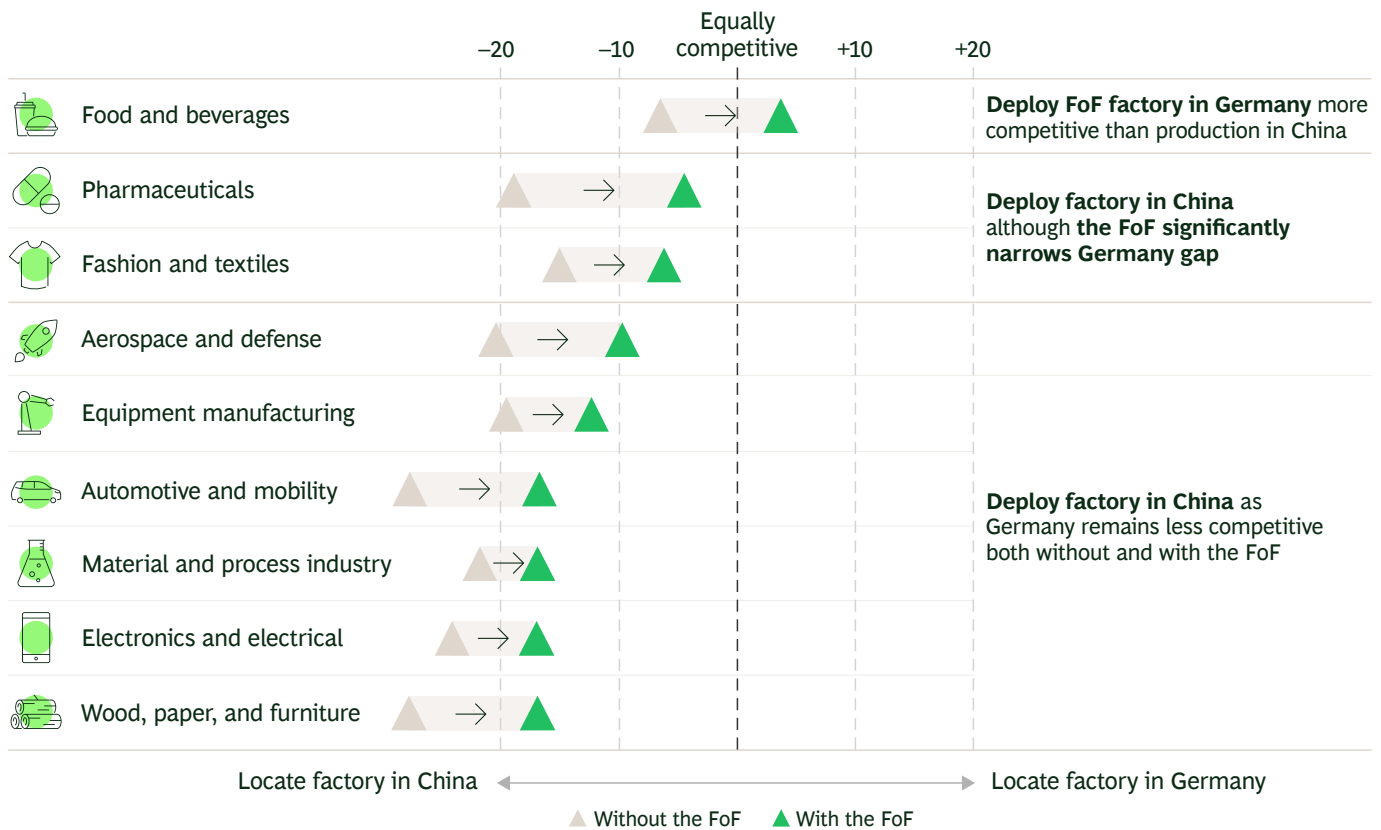
Americas. Tariff-driven trade realignments are creating short-term uncertainty—most companies have deferred strategic decisions on production location waiting for clarity that has yet to materialize. Beyond these immediate dynamics, investing in the FoF and its qualitative enablers will be the decisive factor for long-term manufacturing competitiveness across the region.

EXHIBIT 7

The FoF Advantage Is Less Pronounced for Greenfield Projects

Germany vs. China for greenfield project serving German market

MANUFACTURING COMPETITIVENESS INDEX SCORING GAP (%)



Source: BCG Institute Manufacturing Competitiveness Index.

Note: FoF = Factory of the Future. Manufacturing Competitiveness Index rates on a scale from 1 (very low) to 10 (very high) the weighted combination of 42 cost and qualitative factors in the context of a greenfield German factory serving German market, weightings derived from a survey of 1,000 manufacturing executives.

North America. Tariffs and other trade measures in the US have strengthened the case for local production to serve the domestic market, but only when manufacturers do not have free-trade access to lower-cost countries such as Mexico. If such access exists, manufacturers may choose to localize production in Mexico rather than reshore meaningfully to the US. At the same time, because retaliatory trade measures erode export competitiveness, the overall effect of tariffs is mixed, even if they encourage some domestic-market localization. This creates a structural tension: protection can reduce the immediate pressure to invest in the FoF, leaving US factories less competitive than peers elsewhere. If other regions continue to scale advanced manufacturing capabilities, protected industries risk falling behind, with higher local prices and greater exposure if trade barriers ease over time.

Investing in the FoF changes that equation by building long-term manufacturing competitiveness in sectors such as automotive, where the combination of automation and proximity to demand can tip the balance durably in favor of domestic production. In other sectors, such as fashion and textiles or electrical and electronic components, nearshoring alternatives to Mexico are likely to retain a strategic role.

South America. As manufacturing networks reconfigure to serve North American markets, South America emerges as a diversification alternative to Mexico—particularly for labor-intensive sectors such as fashion and textiles, where South American countries such as Peru and Colombia can offer meaningful cost advantages. Yet maintaining competitiveness in an increasingly FoF-driven world will require sustained investment in the qualitative dimensions that matter most: digital infrastructure and workforce skills. Without that foundation, the region’s ability to capture a growing share of footprint opportunities—and move up the value chain over time—can be constrained.

Europe. The FoF offers a significant but differentiated opportunity across the continent—acting as a protective play in some sectors for manufacturers based in Western Europe, while reinforcing Eastern Europe’s role as a strategic nearshoring alternative.

Western Europe and the Nordics. For advanced manufacturing economies in Europe and the Nordics, the FoF can materially strengthen competitiveness in sectors serving domestic or regional markets—particularly for existing brownfield factories. Industries such as food processing, defense, and automotive benefit from a FoF-driven narrowing cost gap in lower-cost regions, combined with proximity to customers and strong qualitative readiness on skills and digital infrastructure. However, the FoF does not close every gap. In other strategic sectors—such as electrical manufacturing (batteries, semiconductors)—structural cost disadvantages remain significant even with advanced automation. And in some locations, high labor restructuring costs can make FoF productivity gains more costly and slower to achieve, tempering the pace of adoption.

Eastern Europe. Eastern Europe has become a nearshoring alternative to Asia for Western European manufacturers. This is particularly the case in automotive, equipment manufacturing, and electrical components, with Poland, Hungary, and the Czech Republic as key manufacturing hubs. Yet the region faces intensifying competition from South and Southeast Asia, notably in labor-intensive sectors. Investing in the FoF would strengthen its nearshoring role—especially in strategic sectors where Western Europe is structurally less competitive, such as electrical equipment and pharmaceuticals. Sustaining that position will nonetheless require continued investment in digital infrastructure and workforce skills.

Africa and the Middle East. This region has emerged as a strategically important manufacturing base for serving high-cost regions. For instance, Morocco has established itself as a nearshoring alternative for Europe, particularly in automotive supply chains. The UAE is a competitive location for energy-intensive industries such as steel, aluminum, and chemicals, leveraging low energy costs and a rapidly improving business environment and industrial infrastructure. The FoF reshapes both positions: automation risks eroding the labor cost advantages that have driven the region’s appeal—particularly for Morocco relative to Eastern Europe as a nearshoring hub for Europe—while also creating an opportunity to consolidate its current position, provided investment in workforce skills and digital infrastructure accelerates. The UAE is already charting this path by investing massively in its AI capabilities to support advanced manufacturing.

Asia. While Asia is the most cost-competitive manufacturing region today, the FoF is reshaping Asia’s competitive landscape. It is strengthening China’s position in some sectors while creating new opportunities for East Asian economies. The FoF is also raising the stakes for South and Southeast Asian economies to move beyond their current labor cost advantage before it is diminished by automation.

China. China is currently highly competitive across a broad range of industrial sectors, combining cost advantages, scale, supply chain depth, and extended access to critical raw materials. While China outcompetes high-cost regions like Europe across most sectors, it is also increasingly challenged by South and Southeast Asian economies—notably India and Vietnam. These countries are emerging as viable alternatives in labor-intensive sectors such as textiles, electronics, and vehicle components, offering manufacturers a diversification option to reduce dependence on Chinese supply chains.

The FoF represents a significant opportunity to defend China’s competitive position. While automation reduces China’s cost advantage over high-cost regions—especially for sectors where logistics, tariffs, and proximity to demand play a larger role, such as food—it sustains China’s edge in others. This is especially notable in equipment manufacturing and electrical and electronics equipment. In addition, the FoF strengthens China’s competitive position against South and Southeast Asia by narrowing the labor cost gap. China is already seizing this opportunity by investing massively in FoF technologies to maintain its central role in global manufacturing networks.

Japan and South Korea. Both locations face growing competitive pressure as they are challenged by China and South and Southeast Asian countries in sectors such as automotive, equipment manufacturing, and electrical products. The FoF offers a path to reverse this declining competitiveness: by combining FoF automation potential with Japan's and South Korea's skilled workforces and robust digital infrastructure, these locations can reclaim competitiveness in these sectors and emerge as alternative hubs to China for pharmaceuticals and high-value electronics. Both countries are already moving decisively in this direction, with South Korea recording the world's highest robot density in manufacturing, according to the International Federation of Robotics.

South and Southeast Asia. India, Vietnam, and other South and Southeast Asian economies have emerged as the primary beneficiaries of supply chain reconfiguration in Asia. They offer a cost-competitive alternative to China in labor-intensive sectors such as textiles, electronics assembly, and select automotive components, to serve global markets. While the FoF narrows this labor cost advantage, it also represents an opportunity: investing now would allow these locations to secure long-term competitiveness, anticipating potential labor cost increases that would gradually erode their current edge—while also meeting the production quality and consistency that global market standards increasingly demand. India is already moving in this direction: auto component exports are targeted to reach \$100 billion by 2030, more than two-thirds of manufacturers in this sector are at **FoF pilot stage or beyond**, and more than one-third are actively scaling FoF initiatives.

Making Decisions in a New Manufacturing Landscape

For manufacturers, long-held assumptions about cost and location are being upended. AI is changing what production setups can achieve, tariffs are redrawing the global map of trade, and geopolitical volatility is forcing leaders to walk a fine line between efficiency and resilience.

Where to produce can no longer be assessed independently of how production is designed. Instead, competitiveness depends on the ability to deploy advanced manufacturing operations, achieve their productivity benefits, and align those capabilities with the broader footprint. Because technology, adaptability, and system-level productivity increasingly dictate where value can be created most effectively, CEOs must consider their footprint strategy and technology investments in tandem.

For leaders, this creates a more complex and urgent decision landscape but also one that is more flexible. There is no single right answer: in some contexts, redesign will favor upgrading existing operations; in others, relocation will remain the better option. The advantage will go to companies that treat footprint and technology as a single, integrated decision.

In the coming decade, the competitive frontier in manufacturing will be defined by which manufacturers successfully redesign their production systems—and where they choose to deploy them.

Interested in learning more? Explore the Factory of the Future with our interactive tools at [BCG.com](https://www.bcg.com).



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